Dharmik Umeshkumar Jha

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Portfolio

EDUCATION

Nirma University, Institute of Technology, Ahmedabad, Gujarat, India. *2023 – 2027*

- **Major:** B.Tech Mechanical Engineering (in progress) — GPA: 8.49/10.0
- **Minor:** Data Science
- **Relevant Coursework:** Machine Design, Finite Element Analysis, Design of Machines, Dynamics, Aerodynamics, Automation & Control, Manufacturing Processes

EXPERIENCE

Society of Automotive Engineers (SAE) Nirma Collegiate Club — Team Arrow *Oct 2023 – Mar 2026*

Core Member (2023) → Design Lead (2025) → Captain (2025–2026)

- Directed a 9-member team to **1st Overall** at SAE DDC 2025 (100+ teams); led full design-build-test cycle of a fixed-wing UAV with <4 kg empty weight, 6 kg payload capacity, and a 500 g airdrop mechanism — managing configuration, structural sizing, and systems integration across a 5-month cycle.
- Captained 9-member team to design and fabricate a hybrid tilt-rotor tricopter UAV (800 mm wingspan, 1.6 kg AUW) for SAE Aero Design East 2026; aircraft integrates a custom Autonomous Ground Vehicle (AGV) payload system projected to score 52 competition points under a 4-minute autonomous mission profile.
- Led design of a sub-2 kg quadrotor (440 mm wheelbase, 1.847 kg final AUW) for Aerothon 2025; applied Fusion 360 generative design to consolidate frame hub and rotor arms into a single PA12 monocoque, achieving a thrust-to-weight ratio of 3.86 and 21.57-minute hover endurance.
- Contributed to aerodynamic analysis, structural sizing, and composite fabrication of fixed-wing UAVs across 3 national and 2 international competitions; SAE Aero Design West 2025 entry achieved **11th overall** and **5th in mission score** among 75 international teams.

RESEARCH PUBLICATION

Conference Paper

- *Design Evolution and Aerodynamic Performance Assessment of Regular-Class Fixed-Wing Unmanned Aerial Vehicle (UAV).* Aerodynamic analysis performed using XFLR5; computational fluid dynamics validation conducted in collaboration with research team. Presented at FMFP 2025 (12th International & 52nd National Conference on Fluid Mechanics and Fluid Power), Nirma University, December 2025.

Journal Manuscript (In Preparation)

- *Aerodynamic Performance Assessment of High-Payload Fixed-Wing UAV.*
  Manuscript in preparation for submission to the Journal of Flow Visualization and Image Processing.

SIGNIFICANT ACADEMIC PROJECTS

Design and Development of Advanced Class UAV – SAE Aero Design East 2026 *Sept 2025 – March 2026*

- Designed tilt-rotor tricopter (800 mm wingspan, 1.6 kg AUW, S1223 + NACA 9325 lifting-body fuselage) capable of autonomous hover-to-cruise transition; front two thrust vectors rotate 110° via KST X10 Pro A servo (12.5 kgf·cm torque) to achieve stable fixed-wing flight at 16 m/s cruise.
- Sized tri-motor T-Motor F80 Pro 2500KV propulsion system to deliver thrust-to-weight ratio of 2:1; static bench testing confirmed 3.6 lbs thrust at 50% throttle, providing a 36% thrust margin above hover requirement.

Aerothon 2025 – Disaster Response Quadrotor UAV *March 2025 – August 2025*

- Designed a sub-2 kg quadrotor (440 mm wheelbase) for disaster response applications. Developed a single integrated airframe using **generative design** to optimise strength-to-weight ratio and manufacturability. Fabricated using 3D-printed PETG for lightweight structural reliability.

Design of Micro Class Aircraft – SAE Aero Design West 2025 *Sept 2024 – April 2025*

- Designed, fabricated, and tested a 1.6 kg micro-class UAV achieving a payload fraction >0.5 under a 450 W power limit.

Achieved short takeoff (<10 ft), 5-minute endurance, and high structural integrity through aerodynamic optimisation and validation testing. Aircraft competed at SAE Aero Design West 2025 (California, USA) — achieved **11th place overall** and **5th in mission score** among 75 international teams.

Design of Regular Class Fixed Wing Aircraft – SAE Drone Development Challenge *Oct 2023 – March 2025*

- Designed, fabricated, and tested a fixed-wing UAV with empty weight <4 kg, optimised for high payload capacity within geometric constraints (wingspan <72 in; total dimensions <150 in). Constructed using balsa and aeroply with aluminium and acrylic spars for enhanced rigidity. Integrated a $10'' \times 4'' \times 4''$ payload bay with 500 g airdrop capability in conventional tail configuration.

Design of Primary Landing Gear – SAE Drone Development Challenge *Oct 2023 – March 2025*

- Designed a lightweight primary landing gear for a 10 kg fixed-wing UAV using carbon-fibre rods secured with CNC-machined aluminium clamps. Integrated an expanding spring mechanism for landing impact absorption. Final assembly weighed $\sim$250 g — representing less than 2.5% of total aircraft weight — while sustaining rated landing impact loads validated through drop testing.

TECHNICAL SKILLS

- **Design & Simulation:** SolidWorks, Fusion 360, ANSYS (Structural), XFLR5
- **Programming & Data Analysis:** Python (NumPy, Matplotlib), MATLAB
- **Manufacturing & Prototyping:** 3D Printing, Laser Cutting, Composite Fabrication
- **Flight Systems & Integration:** Pixhawk Autopilot, Mission Planner

HONOURS & AWARDS

- Achieved **overall 11th place** and **5th place in mission score** out of 75 international teams at SAE Aero Design West (Micro Class) 2025, Apollo XI RC Field, Van Nuys, California, USA — April 2025.
- Secured **1st position** in the Regular Class Category and **3rd position** in the Micro Class Category at the SAEISS Drone Design Challenge 2025, Karpaga Vinayaga College of Engineering & Technology, Chennai — March 2025.
- Achieved **overall 4th place** and **2nd Best Design Report** in the Regular Class Category at the SAEISS Drone Design Challenge 2024, SRM University, Chennai — September 2024.